

Only Testing Will Tell

Substantial differences result from performing a Duct Blaster test during the framing stage (left) and at completion (right). A battery of tests pinpoints why.

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Quantifying A/C duct leakage and identifying when and where it takes place can make a big contribution to energy efficiency. Consequently, duct leakage testing is a major part of the diagnostic and inspection work performed by D.R. Wastchak, L.L.C., in Phoenix, Arizona. Since January 1999, we have completed more than 1,500 duct leakage tests and a similar number of blower door tests on new homes. Approximately 90% of our compliance/quality control testing of A/C systems is on homes at the framing stage of construction, just after the systems are roughed in; however, there are many instances when we must test new systems after a home is completed and the A/C system is fully operational. We observed that test results at this latter stage of construction were consistently higher, showing leakier A/C systems, than results of tests done during the framing stage. This led us to wonder if there was a predictable difference that would allow us to extrapolate from one test result to the other, depending on when the leakage data were collected.

To evaluate these differences, we developed a battery of tests on a sample of homes in the Phoenix market (see Table 1). The data we obtained enabled us to quantify leakage at air handler units, at metal supply boots (to outside), and at supply registers (to inside). Furthermore, we were indeed able to



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extrapolate the leakage at air handler units, boots, registers, and ducts, based on duct leakage data collected when a house is completed. This procedure greatly enhances our ability to locate problems in a new system. Finally, we used the test battery to study the differences between measurements of duct leakage using a Duct Blaster and measurements using the blower door subtraction method.

Completing the Battery of Tests

Our results are based on 29 houses tested during 1999–2000 in the Phoenix area. These homes were built by three different builders and were located in 11 different communities. They ranged in size from 1,058 to 3,050 ft². In total, 36 separate heating and A/C systems were tested (some larger homes had two

units), ranging in size from 2.5 to 5 tons. All homes had split systems with the air handler units (AHUs) located in the unconditioned attic and the condensing units located on the ground outside the home. The air distribution systems were 100% flex duct (supply and return), also located in the unconditioned attic, with a single central return located in the ceiling not far from the AHU. Thirty-two of the systems had a furnace–air conditioner setup and the remaining four were straight heat pumps. The same A/C contractor installed all of the A/C systems in the test battery.

We first tested each home in our sample during the framing stage of construction with a Minneapolis Duct Blaster. When performing a Duct Blaster test, each of the supply outlets is covered and the Duct Blaster fan is attached to the return side of the system. With the entire system sealed, the volume of air drawn out of the system (in depressurization mode) is equivalent to the air leaking into the closed system. In all duct leakage testing, our results included leakage at return ducts, supply ducts, and the AHU. We used a flow conditioner with the appropriate low-flow rings during all depressurization tests, in accordance with the manufacturer's operating instructions. The Duct Blaster uses a digital gauge to automatically calculate the leakage rate for the system, measured in CFM, at a pressure of 25 Pa (CFM₂₅). We performed the Duct Blaster test after the air handler and ducts were completely roughed in. At this stage of construction, we were able to get a very good seal on the metal boot boxes (the termination point of each supply duct) with our grille mask tape. The same was true at the metal return boxes. Note that testing was completed on each system with the AHU as installed by the A/C contractor and again with the holes in the AHU cabinet sealed as tightly as possible with tape.

When each home was fully completed, we again tested the heating and A/C systems with a Duct Blaster. We tested each system with the supply registers taped completely to the

drywall (complete-taped) and taped only on the face of the register (face-taped). Next, we used a Minneapolis Blower Door to complete a subtraction method test to measure duct leakage for each home (in the face-taped condition). For the subtraction method, two measurements are taken: one with the ducts sealed, the other with the ducts open. The difference between the two measurements provides the duct leakage to the outside at the given test pressure. The blower door subtraction test was completed at the standard 50 Pa blower door test pressure and again at 25 Pa, so that we had leakage values comparable to those for the Duct Blaster. Note that the blower door subtraction method underestimates actual duct leakage, as our data demonstrated (see "Subtraction Method vs. Duct Blaster Leakage to the Outside," p. 27).

The final test in the battery was duct leakage to the outside of the building envelope, measured at 25 Pa, using the Duct Blaster and blower door simultaneously.

Table 1. Summary of Battery Tests

Test Name	Equipment	Stage of Construction/Description
1) Frame-stage leakage —unit untaped	Duct Blaster	HVAC system rough-in complete, holes in AHU untaped
2) Frame-stage leakage —unit taped	Duct Blaster	Test 1 again with holes in AHU taped to seal the unit
3) Finish-stage leakage —face-taped	Duct Blaster	House and HVAC system completed, unit untaped, face of supply registers taped
4) Finish-stage leakage —complete-taped	Duct Blaster	Test 3 again with supply registers taped to drywall
5) Subtraction method	Blower door	Blower door run first with supply registers face taped, then with the tape removed and ducts open; unit untaped
6) Leakage to outside	Duct Blaster & Blower door	Both pieces of equipment run simultaneously with ducts open, unit untaped

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Air Handler Leakage

One of our goals was to determine the AHU's contribution to total leakage. This is useful for performance testing, since it isolates leakage at the AHU (which the contractor has no control over); any remaining leakage is due primarily to holes in the duct system itself. There are also two reasons why it is important to know air handler leakage during house operation. First, the system pressures are the highest at the air handler; a leak of a given size would have the largest impact there. Second, the leaks at the air handler often bypass the filters.

Average AHU leakage did not change much with various system sizes—approximately 17 CFM₂₅ (see Figure 1). However, the battery test data indicated a fair amount of variation from the average. AHU leakage was mostly in the 10–25 CFM₂₅ range.

Using the battery test data for AHU leakage and leakage measured when the houses were completed, we calculated

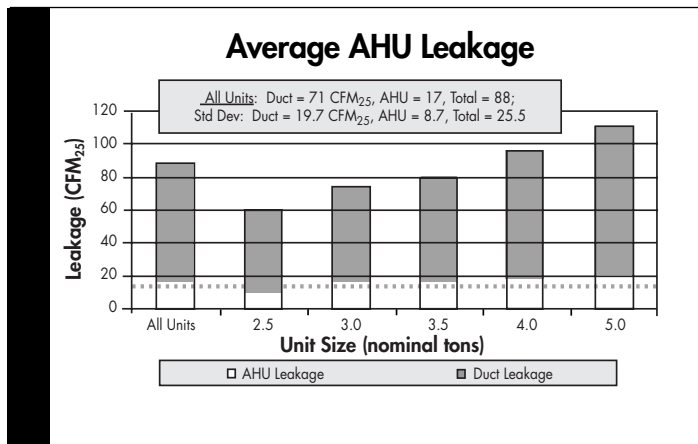


Figure 1. Leakage from the air handler units was primarily in the 10–25 CFM₂₅ range

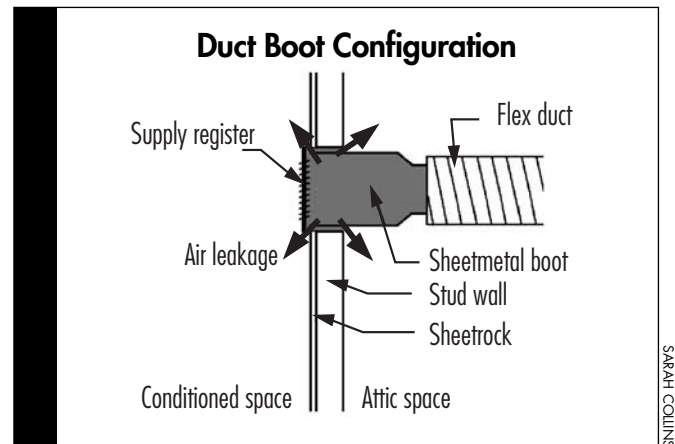


Figure 2. If the boot is not completely sealed or caulked to the drywall, leakage to the inside and/or the outside can occur.

an average ratio of AHU leakage. The test data at completion were measured with the face of the register taped (face-tape leakage). The average ratio of AHU leakage to face-tape leakage was 10% ($\pm 5\%$). For instance, if the face-tape leakage test was 100 CFM₂₅, and the AHU leakage was 10 CFM₂₅, the ratio would be 10%.

It is interesting to note that in a sample of 253 duct leakage tests that Wastchak conducted earlier, the data spread was wider. Although the data ranged from 4 to 184 CFM₂₅, AHU leakage was mostly in the 5 to 45 CFM₂₅ range. Average leakage for this data set was 31 CFM₂₅. Because these data were derived from daily field testing rather than from the test battery conducted for this study, the reason for the variation from the average is not known.

Building Science Corporation (BSC) and Guaranteed Watt Saver (GWS) both conducted studies on the leakage of isolated AHUs (units with no duct system attached) in a shop condition. BSC found leakage rates from 45 to 60 CFM₂₅ with an average of 54 CFM₂₅ (standard deviation = 6 CFM₂₅). GWS leakage data ranged from 28 to 69 CFM₂₅ with an average of 48 CFM₂₅ (standard deviation = 15 CFM₂₅) (see Table 2). Both BSC and GWS studies measured leakage with and without the cooling coils and plenums. The 54 and 48 CFM₂₅ averages are for

measurements with the coils and plenums in place, which is comparable to the battery and large sample test conditions described above. It would be logical that field-measured leakage

This research demonstrated that a reasonable prediction can be made of the relative leakage rates for the components in an A/C system.

results would be lower than shop-measured isolated results since in a field test, sealing the AHU with tape eliminates most—but not all—of the leakage.

Boot vs. Total-System Leakage

One common question in duct leakage testing is, how much does leakage around the duct boot contribute to total-system leakage measurements? If the boot is not completely sealed or caulked to the drywall, leakage to the inside and/or the outside can occur (see Figure 2). The test battery gave us an ideal opportunity to find some quantitative answers to this question.

In order to estimate leakage at the boots, we subtracted the leakage measured during the framing stage of

construction (that is, sealing directly to the roughed-in metal boot can) from face-tape leakage (that is, leakage measured at completion, with just the face of the register taped). The difference represents boot leakage alone, to both the inside and the outside, assuming that the frame-stage leakage test sealed the boots completely, driving all leakage to joints in the duct system, and that no leakage was added to the system between the framing stage and completion (that is, there were no damaged ducts).

Leakage rates varied widely—from 7 to 136 CFM₂₅. These results are to be expected, because the numbers are not normalized based on the floor area, the number of registers, or the amount of total leakage in each home. The average ratio of frame-stage leakage to face-tape leakage was 45% ($\pm 11\%$), or approximately 35% to 55% (see Figure 3). Knowing this ratio, one could also estimate that approximately 45% to 65% of face-tape leakage is from the remainder of the duct system and AHU.

Leakage to Inside at Registers

Almost all of the testing that we conduct in completed homes is done with only the face of the register covered (versus taping all the way to the drywall). We do this in order to avoid touching newly painted walls and, even more importantly,

Subtraction Method vs. Duct Blaster Leakage to the Outside

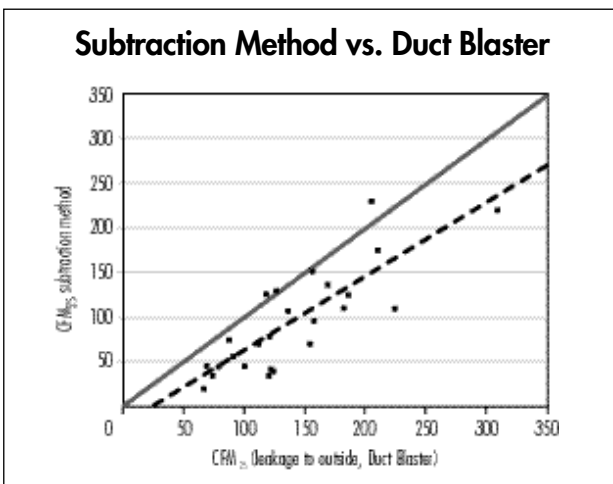
Although Duct Blaster tests are generally considered to be the most accurate method of measuring duct leakage, it is sometimes necessary to use alternate methods due to budget, time, or equipment constraints. One typical technique to measure duct leakage to the outside is the subtraction method, using a blower door. Two measurements are taken: one with the ducts sealed, the other with the ducts open. The difference between the two measurements provides the duct leakage to the outside at the given test pressure.

Again, the test battery provided an excellent opportunity to compare Duct Blaster and blower door subtraction method duct leakage results. The subtraction test results are graphed against the duct blaster leakage to outside test results (see Figure A). Note that in houses with two air handlers, the sum of the leakage to the outside values is compared to the subtraction result. Both tests were run with the face of the register taped so as to focus only on leakage to the inside. The ideal result (identical measurements from both methods) is shown by the solid line.

We found some correlation ($R^2 = 0.6761$) between the Duct Blaster and the subtraction method results. The Duct Blaster results varied from 67 to 309 CFM₂₅, with an average of 140 CFM₂₅. In comparison, the subtraction measurements varied from 20 to 230 CFM₂₅, with an average of 95 CFM₂₅.

The battery test data show that the subtraction method can result in substantial errors. The average error was 37% (the difference between the two measurements divided by the Duct Blaster test). This error ranged from 2% (very close) to 71% (35 CFM₂₅ vs. 120 CFM₂₅). The bottom line is that you might get good data with a subtraction test, but you don't know for sure when it happens.

There are several reasons for the errors produced when measuring duct leakage by the subtraction method. Perhaps the most significant reason for error is that the



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Figure A. The average subtraction-method leakage was 95 CFM₂₅, compared to 140 CFM₂₅ with the Duct Blaster.

blower door is taking the difference between two large whole-house leakage air flows. This means that a small percentage error in testing will produce a large error in duct leakage. In addition, the subtraction test assumes that once the ducts are masked, there is no connection at all between the house and the ducts—which is seldom the case. When there is a connection, the subtraction test result will underestimate leakage to the outside. This situation is shown in Figure A (data is below the solid line).

A calculated subtraction correction factor (SCF) can be used to correct this error. The SCF is a multiplier applied to the test that is selected based on the pressure difference between the house and the ducts during the sealed-duct test. With careful use of the SCF, the accuracy of the subtraction method can be increased substantially. The table of SCF values, as well as an extensive discussion of the subtraction method for measuring duct leakage, are available on the Energy Conservatory's Web site. See www.energyconservatory.com/usertips.html, "User Tip #3: Using the Blower Door subtraction method for estimating duct leakage."

to avoid peeling paint off the wall when removing the tape. Leakage at the face of a register is to the inside of a house, that is, back into the conditioned space. This leakage is much less problematic than leakage to the outside because it is more or less helping to heat or cool the house.

We found a wide spread of data for complete-tape minus face-tape leakage. Again, this wide spread (5–78 CFM₂₅) is to be expected, because it reflects only the difference without any normalization. The average ratio of complete-tape leakage to face-tape leakage

was 80% ($\pm 8\%$) (see Figure 4). As in the case of boot vs. total system leakage, knowing this ratio, one could also estimate that approximately 10% to 30% of the leakage is from around the face of the registers.

Using Ratios to Calculate Proportional Component Leakage

The battery test data provide a means for estimating the various component

leakages in an A/C system if testing can be done only after the house is completed, which is often the case in diagnostic testing. In the duct blaster tests shown (face-taped), we estimate that 10% of the leakage is attributable to the AHU. Likewise, we estimate that 20% of the boot leakage is to the inside around the face of the registers. Through extrapolation, we can further estimate that 25% of the total measured leakage is to the outside at the boots and 45% is attributable to the ducts (see Figure 5). See a "Practical Application" of these ratios, p. 29.

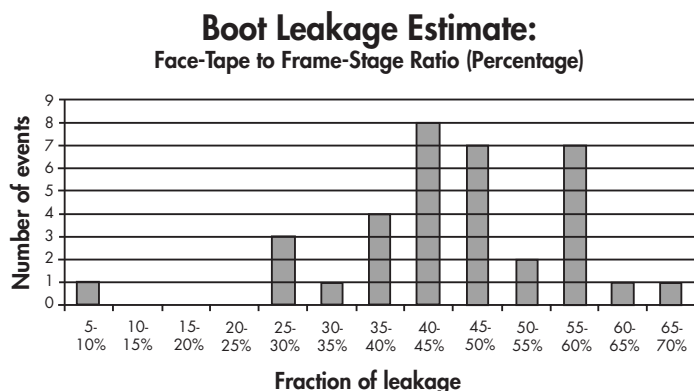


Figure 3. Boot leakage is estimated at 35% to 55% of total leakage.

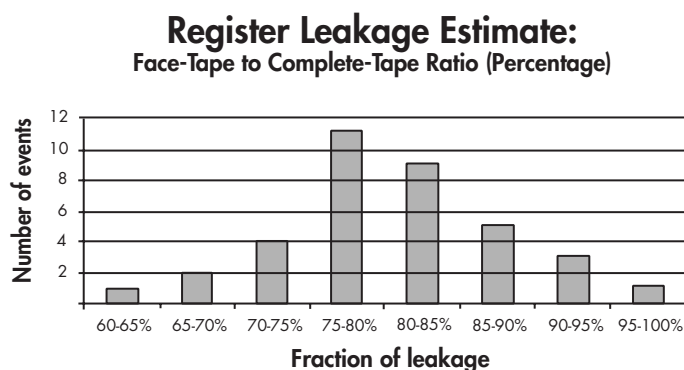


Figure 4. Boot leakage to the inside is estimated at 10% to 30%.

Table 2. Summary of Results from Multiple Studies

Study	Average AHU Leakage (CFM ₂₅)	Range (Measured)	Range (±1 St. Dev.)	Number of Samples
1.) Battery testing	17	5 - 40	8 - 25	31
2.) Large DRW field sample	31	4 - 184	8 - 54	253
3.) Building Science Corp.	54	45 - 60	48 - 60	5
4.) Guaranteed Watt Saver	48	28 - 69	47 - 63	6

Note: Studies 1 and 2 are based on field data from taped and untaped AHU's installed in homes; studies 3 and 4 are based on data collected in a laboratory/shop setting. Studies 1 through 4 all measured leakage in AHUs with coils and plenums included.

Proportional Leakage Ratios
(Face-Tape Testing at Completion)

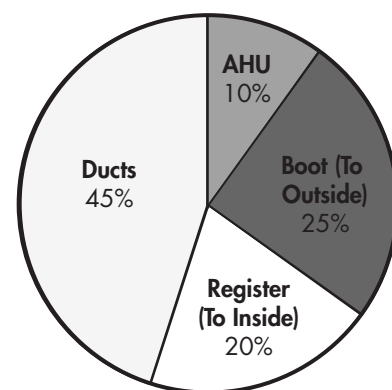


Figure 5. Total leakage percentages are apportioned based on the results of the battery of tests.

Note that these estimates are based on test results for flex duct sealed with mastic, and for AHUs and ductwork located entirely in the attic. Furthermore, these results are from one city during one time period, with ductwork and AHUs installed by one HVAC contractor in new construction. We do not recommend using these results to calculate estimates in existing homes that are known to have significant failures or installation problems (i.e., in homes that may be tested for possible retrofit). Leakage in these homes is likely to be greater than leakage caused by common problems in new construction, such as failing to properly seal a boot, Y connection, junction box, or metal plenum collar. Finally, the standard deviations noted with each set of data will have a significant impact on the accuracy of any prediction. That said, we feel that our estimated ratios are applicable to A/C systems that are installed in a similar fashion to systems installed in the Phoenix market, and in locations such as Tucson, Nevada, Oklahoma, Texas, Southern California, and the Southeast.

Reasonable Predictions

This research demonstrated that a reasonable prediction can be made of the relative leakage rates for the components in an A/C system. Although we recognize the limi-

Practical Application

Knowing the proportional leakage ratios derived from the battery test data, it is possible to apply the average ratios to duct leakage test results in the field to estimate the proportion of leakage attributable to registers, boots, AHUs, and ducts. The following example demonstrates a practical application of these ratios.

Assume that a home was tested with a Duct Blaster when completed and that the faces of the registers were taped for the test. If the face-taped test result was 210 CFM₂₅, the calculations shown below can be used to estimate leakage at the various components.

Proportional Leakage Estimator

Measured leakage (face-tape at finish):

210 CFM₂₅

Leakage Estimates:

Leakage at Register (to inside):	20%	x	210 CFM ₂₅	=	42 CFM ₂₅
Leakage at Boot:					
Total	45%	x	210 CFM ₂₅	=	95 CFM ₂₅
To Outside	95 CFM ₂₅	-	42 CFM ₂₅	=	53 CFM ₂₅
Leakage at ducts & AHU:	55%	x	210 CFM ₂₅	=	115 CFM ₂₅
Leakage at AHU:	10%	x	210 CFM ₂₅	=	21 CFM ₂₅
Leakage at ducts:	115 CFM ₂₅	-	21 CFM ₂₅	=	94 CFM ₂₅

Percentage of Face-tape Leakage

AHU	10%	21 CFM ₂₅
Boot (to outside)	25%	53 CFM ₂₅
Register (to inside)	20%	42 CFM ₂₅
Ducts	45%	95 CFM ₂₅
	100%	210 CFM₂₅

tations of a set of ratios when it comes to predictability, we feel that our findings provide a guideline for how leakages are distributed within a new A/C system. They will give the A/C contractor a better feel for the proportion of leakage over which their ductwork installers have control.

Considering our initial point of departure, namely determining why duct leakage results at frame differed from leakage results at finish, we now know that this difference is attributable primarily to the leakage at the boots and we can estimate this leakage at 20% around the face of the register and 25% back into the attic. This is based on the assumption that leakage at the AHU and ducts has not changed from the frame-stage to the finish-stage of construction.

We have made the HVAC contractors in Phoenix aware of this leakage at the boots, and now contractors are beginning to demand that manufactur-

ers send the sheet metal boots to the field pre-sealed with mastic. This movement is driven, in part, by the need to reduce leakage in any way possible to meet the stringent leakage requirements of the Engineered for Life and Environments for Living programs. We hope that the manufacturers of AHUs will take notice of their contribution to A/C system leakage, and that they too will make the necessary modifications to their products.



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Colby Swanson (formerly with D.R. Wastchak, L.L.C.) conducted the field testing and data collection for this research. The DOE Building America program provided financial assistance for this research.

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